

A cubic zero-free boundary from plain-moment parameter feedback

RH–Weil research project

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A derivation relative to explicitly imported results

Abstract

We extend the usable parameter range of the generic plain moment in the September 30, 2026 preprint *The Quasi-Riemann Hypothesis*, by replaying its proof from its specified underlying inputs. The range $37/50 \leq \kappa \leq 1$ retains the original moving conductors, natural zero extensions, coefficient classes and uniform mesh. Coupling the actual moment parameter to the whole-family zero supremum yields the strict zero-free half-plane

$$\operatorname{Re} s > \sigma_* = 0.874957019420098946\dots, \quad 7884\sigma_*^3 - 18819\sigma_*^2 + 14643\sigma_* - 3686 = 0,$$

with the root specified by an exact rational isolating interval. The conclusion is for finite-order Hecke functions over $\mathbb{Q}(\sqrt{-3})$ and, by the imported quadratic transfer, all Dirichlet functions, including zeta. We prove a continuous quadratic certificate at a reference parameter and pay the cost of using the larger actual parameter. We retain the complete low Gram estimate and verify strict detector widths, every physical row range, Euler domains, the principal residue and uniform family continuation. This improves the preceding free- b boundary by approximately $5.04 \cdot 10^{-8}$. The theorem assumes an explicit imported package and its Hecke seven-eighths bootstrap; it is not an independent certification of those inputs or of an external proof kernel. The principal pole is allowed and the boundary line is excluded. No new critical-line zero proportion is asserted.

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1 Statement, source and hypotheses

Let $F = \mathbb{Q}(\sqrt{-3})$. We use the ideal and element representatives, norm q_a , units, ray conditions and zero-extension conventions of the pinned source [1]. A superscript S deletes the finite Euler factors in the final fixed arithmetic datum. This paper is separate from the papers giving 69999/80000 and the free- b boundary [2, 3]; those delivered artifacts are preserved. The new step is a moment-domain extension and parameter feedback, rather than another choice of b in the old $\kappa = 3/4$ envelope.

Definition 1.1 (Imported package $\mathcal{R}$). We assume the following source results, with their complete coefficient classes, masks, uniformities and side conditions: finite Gauss and full-correlation identities, reciprocity, complete support and natural zero extensions; coefficientwise probe and Poisson identities, finite compensation, local arithmetic tables and ray calibration; smooth separation, Gaussian annuli, reflection sectors, generic reflected energy and the full quantitative additive Gram bound; global logarithmic and buffered reciprocal control, primitive functional equations, growth and complete contour tails; detector witnesses, marked and recursive inverse moments and sixth-power amplification; fixed-field counting and fixed-ray prime asymptotics; the whole finite-order Hecke seven-eighths zero-free theorem; and the quadratic factorization transferring that theorem to Dirichlet functions. The new plain-moment range below is derived from the underlying identities and estimates, not assumed by extending the statement of the original plain theorem outside its published range.

Define

$$\beta_* = \sup(\{1/2\} \cup \{\operatorname{Re} \rho : L_F(\rho, \eta) = 0, \rho \text{ nontrivial, } \eta \text{ primitive of finite order}\}).$$

The principal pole is excluded. The only bootstrap used in the proof of the new boundary is $\beta_* \leq 7/8$ from $\mathcal{R}$. The previous free- b boundary is used only for comparison. Pure norm twists change imaginary parts, not this real supremum.

The source is frozen at commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. Its September 30 `build/paper.tex` has canonical LF SHA-256 `42a5ee0febca59fd1def55cfd6c6808c322ef4237d7726303ea52f711deac6a3`. Line ranges below refer to that file. Table 1 specifies the interfaces used, so that the relative nature of the result is explicit.

Table 1: Imported interfaces and retained conditions.

Interface	Source lines	Use in this paper
Smooth, Gaussian and global log control	1123–1209, 1288–1315, 1536–1594	Joint density, all heights; the new real strip is compact inside the original global domain.
Actual detector and marked moments	4220–4685, 9220–9269	Original presentation, witnesses, coefficient group and strict widths.
Generic plain proof	12492–14974	Underlying proof replayed in Section 2; no out-of-range invocation of its original conclusion.
Physical slots and counts	14985–15446, 15920–15992	Whole-slot mean gain, conjugation, masks, inducing exceptions and shared mesh; new crossing derived below.
Reflection and Gram	7663–7804, 8340–8364	All frozen parts, actual dyads and complete third Gram term.

Interface	Source lines	Use in this paper
Physical identities and local bounds	8707–9049, 9101–9195	Same finite compensation, normally convergent corrections and full completed sums.
Whole-bin, principal and outer	5808–5896, 6021–6463	Global move before local partition; new parameter domains verified in Section 8.
Uniform choice and continuation	16191–16463, 6520–6580, 400–502	Losses and mesh before slots; target before internal height orders; external tails last.
Dirichlet transfer	6705–6780	Quadratic factorization, finite Euler deletions and principal pole.

Let e be the unique root in $(1/6, 167/1000)$ of

$$p(e) = 657e^3 - 954e^2 + 21e + 20 = 0. \quad (1)$$

It is strictly isolated by

$$\frac{16683858898627}{10^{14}} < e < \frac{16683858898628}{10^{14}}. \quad (2)$$

Set

$$\sigma_* = \frac{11}{12} - \frac{e}{4}, \quad \kappa_* = \frac{5}{6} - \frac{e}{2} = 2\sigma_* - 1, \quad b = \frac{-5181 + 156335e - 387630e^2}{81941}. \quad (3)$$

Theorem 1.2. *Assume $\mathcal{R}$. Every finite-order Hecke L -function over F has no zero in $\operatorname{Re} s > \sigma_*$, with the root specified by (1)–(3). The same holds for all Dirichlet L -functions, including ζ . Principal functions may have their pole at 1. The boundary line is not included.*

Substitution gives the cubic for σ_* in the abstract. The numerical values

$$\begin{aligned} e &= 0.166838588986270882\dots, \\ \kappa_* &= 0.749914038840197892\dots, \\ b &= 0.123405685419541002\dots \end{aligned}$$

are for orientation; the proof uses rational arithmetic and the specified real embedding. We make no claim of optimality over all possible probes or arithmetic estimates.

2 Replaying the plain moment below three quarters

The original generic plain statement imposes $\kappa \geq 3/4$. It is essential to reprove the statement in the enlarged range. We describe every numerical dependency on that lower endpoint; the source identities remain imported inputs.

Proposition 2.1 (Extended plain range). *Use the source normalized plain sums $S_{\psi_k}(n)$, positive prime-slot product Q_{ψ_k} and row family $\mathcal{R}_z$. Retain rows $q_k \ll Z^m$, the complete displayed moving radical Z^q of the fixed twist, and effective width $M = m + q$. Retain the same polynomial-size fixed extra puncture mask, the finite coefficient group Θ , its fixed finite-ray coefficient lists, disjoint underlying prime supports and nonprincipal inducing-family exclusions. For each $\varepsilon > 0$ and fixed bounded length range, choose the numerical mesh $\eta_{\text{mesh}} > 0$ as specified below. Then, for $37/50 \leq \kappa \leq 1$, positive-slot data satisfying*

$$\begin{aligned} \max_i z_i &\leq \eta_{\text{mesh}}, & n_1 + n_2 + 6\kappa z &\leq M, \\ \beta_* &\leq (1 + \kappa)/2 \quad (\kappa < 1) \end{aligned}$$

obey

$$\sum_{k \in \mathcal{R}_z} |S_{\psi_k}(n_1)S_{\psi_k}(n_2)Q_{\psi_k}|^2 \ll Z^{M+\varepsilon}, \quad (4)$$

with the original finite seminorm and polynomial-height factors. For $z = 0$ retain the original bounded nonnegative plain lengths. The numerical mesh can be chosen after the bounded length range and ε , before the slot count, uniformly in this κ interval and all admissible moving moduli and outer labels.

Proof. Write $A = n_1 + n_2 + z$, so the positive-slot budget is $A + (6\kappa - 1)z \leq M$. Natural zeros remain present even when a phase cancels or is six-divisible; changing a phase does not erase a deleted prime from the moving radical.

The prime estimate in source lines 12833–12900 is obtained on the displaced line to the right of $s_\kappa = (1 + \kappa)/2 \geq \beta_*$. Now $s_\kappa \geq 87/100$, within a fixed compact real strip of the global logarithmic estimate. For the same coefficient expansion and nonprincipal exclusions this gives

$$|Q_\omega|^2 \leq CZ^{\kappa z + \varepsilon_1} p_j(\mathbf{W})^a (1 + |\omega|)^t.$$

The contour displacement is the original $2\varepsilon z$; height and derivative orders are paid by the same weighted Fourier measure. At $\kappa = 1$ the absolute prime count suffices. There is no new principal residue.

At the terminal-width step (12925–12938),

$$z \leq M/(6\kappa) \leq 25M/111, \quad \kappa z \leq M/6.$$

The original displayed $z \leq 2M/9$ is replaced, but it has no later use in the proof. The terminal exponent uses only $\kappa z \leq M/6$.

For the same-band comparison $A > 5M/6$ (12947–12980),

$$(6\kappa - 1)z < M/6, \quad 6\kappa - 1 \geq 86/25, \quad z < 25M/516 < M/20.$$

The two comparison bounds consequently remain

$$A_{\text{comp}} \leq 3M/2 - A + 2z + \xi \leq 23M/30 + \xi,$$

$$A_{\text{comp}} + (6\kappa - 1)z \leq 5M/2 - 2A + 2z + \xi \leq 14M/15 + \xi.$$

Thus the original $M/15$ margin and $\xi \leq \rho/30$ still suffice. The strict inequality $25/516 < 1/20$ has slack $1/645$.

The aggregate support boxes, whole-dyad tails, first complete extraction, two Möbius indicators, common characters, p^6 pool, second correlation and child-width decrease in 13095–13875 do not use the old lower endpoint. Their actual child width is $M' = M + J - g - g_2 + t_2 \leq M - \sigma_{\text{drop}}$. The original pool separation $\ell_{\text{pool}} = \sigma_{\text{drop}}/3$ and $\eta_{\text{pool}} < \sigma_{\text{drop}}/6$ is retained.

In the actual clipping and mask-erasure steps (13876–14310), shortening or removing slots does not increase the affine budget because $6\kappa - 1 \geq 0$. The actual defect remains

$$F_{\text{act}} = (A_{\text{act}} - M'_{\text{act}} + (6\kappa - 1)z_{\text{act}})_+ \leq 6(w + \ell_{\text{edge}}) + \delta_{\text{fr},1} + 2\theta_N.$$

The whole-slot removal cost is $\kappa d_z \leq F_{\text{act}}/6 + \kappa \eta_{\text{pool}} \leq F_{\text{act}}/6 + \eta_{\text{pool}}$. Separate actual rectangles, paired shells, zero extensions and all moving conductor factors are retained; there is no extra cost from extending the interval.

The inducing- Θ exceptions, equal-product cancellation, common centering, sixth-power counts, padding and reflection in 14312–14777 are independent of this lower endpoint. The positive-slot condition still gives $A \leq M$. Finally the numerical Lipschitz bound in 14779–14974 requires only $0 \leq 6\kappa - 1 \leq 5$. Keep the original width floor, bands, finite depth and strict drop

$\sigma_{\text{drop}}/2$. Within each band perform uncentered before centered estimates; establish all zero-slot cases before the positive-slot induction. The same finite induction closes with the same terminal cost and comparison margins.

All numerical losses, mesh and support thresholds are chosen before the fixed slot count. The numerical constants above are uniform in the compact κ interval. After fixing slot and profile data, choose the finite internal seminorm and height orders in the original reverse induction order; an external tail order chosen later cannot alter them. This proves the original uniform contract (4) from $\mathcal{R}$ in the stated range. $\square$

The source's full support, correlation and analytic estimates are still assumptions. The finite algebra audit in the appendix does not verify this induction or those imported estimates.

3 Actual inverse and plain counts

Put $\alpha = 5/6$. In nonfloor bins use $\delta = 2a - 1 \in [1/50, 3/4]$ and the source length-weighted mean amplitude q over *all* physical slots. Error and zero slots have gain zero. Let $x = q/\delta \in [0, 1/2]$ and $y = 1/2 - x$. For $37/50 \leq \kappa \leq 1$ define

$$c = (3\kappa)^{-1}, \quad B_{\text{pl}} = 2 - 2cx, \quad D = 3 - (1 + 2c)x, \quad P = B_{\text{pl}}(1 - x). \quad (5)$$

The actual witness has $r + m \geq t - o(1)$, $r \leq t + o(1)$ and $m \leq 1/2 + o(1)$, with its original inverse and plain spikes. Use two copies of the same plain witness; its fourth spike is not replaced by a single squared spike. The resulting count lines are

$$A_I(r) = 1 - \delta\{x + (1 - x)r\}, \quad S_t(r) = 1 - \delta\{cx + B_{\text{pl}}(t - r)\}.$$

Their crossing is

$$r_*(t) = \frac{B_{\text{pl}}t - (1 - c)x}{D}, \quad R_{\text{short}}(t) = 1 - \delta + \frac{\delta P}{D}(3/2 - t).$$

The long branch remains $1 - \delta + (\alpha - \delta)(t - 1)$. Equating the short and long branches gives

$$J = (\alpha - \delta)D + \delta P, \quad t_\kappa = 1 + \frac{\delta P}{2J}, \quad R_{*,\kappa} = 1 - \delta + \frac{(\alpha - \delta)\delta P}{2J}. \quad (6)$$

These formulas preserve the same character and presentation. Witness and physical heights need not coincide; their parameter Sobolev factors and cumulative $T_1/2$ allowance are retained.

For later strict-width checks, the rectangle satisfies

$$D \geq 455/222 > 2, \quad P \geq 86/111 > 3/4, \quad 35/48 \leq J \leq 5/2, \quad 1 < t_\kappa < 3/2,$$

$$r_*(1) \geq 8/13 > 3/5, \quad r_*(3/2) = 1, \quad 1/3 \leq t - r_*(t) \leq 1/2 \quad (1 \leq t \leq 3/2).$$

The inverse request has $z_M = (1 - r)/2 \leq 5/26 < 1/5$; the plain request has $z_P = (1 - 2m)/(6\kappa) \leq 25/333 < 1/5$. On the inverse branch request $z \leq z_M - \nu_0$, for a fixed $\nu_0 > 0$, and retain both marked widths:

$$1 - r - 2z \geq 2\nu_0, \quad 3 - 2r - 8z = 4(1 - r - 2z) + (2r - 1) \geq 8\nu_0 + 1/5 - o(1).$$

On the plain branch request $z \leq z_P - \nu_0$, giving $1 - 2m - 6\kappa z \geq 6\kappa\nu_0$. Capacity-zero neighborhoods, $r \geq 1$ and $m \geq 1/2$ use the original no-slot or amplified endpoints. Thus a physical supply $e/d > 1/5$ with fixed margin suffices. Whole-slot rounding, Θ expansion and conjugation of the witness and all selected factors together remain essential.

4 The same physical probe and the complete low estimate

Set

$$l_x = (1-e-b)/2, \quad l_y = (1-e+b)/2, \quad M = 1-e, \quad h = (1+3e+b)/2, \quad X = Z^{l_x}, \quad Y = Z^{l_y}. \quad (7)$$

Choose fixed even K , equal $\ell_i = e/K$, $P_i = Z^{\ell_i}$ and the original disjoint nonnegative smooth prime windows. Let I_η be the source completed physical probe, and $I_{\eta;D}$ that same expression with $1_{D|cn^3}$ inserted. For $p_J = \prod_{i \in J} p_i$, $p_\emptyset = 1$, retain precisely

$$I_{\eta, \text{mod}}(Z) = \sum_{(p_i)} \prod_i W_i(q_{p_i}/P_i) \sum_{J \subseteq \{1, \dots, K\}} (-1)^{|J|} q_{p_J}^{-3/2} \overline{\eta(p_{J^c})} \cdot I_{\eta; p_{J^c}}(X/q_{p_J}, Y/q_{p_J}, Zq_{p_{J^c}}).$$

All ray data, annular norm ratios, masks and windows are unchanged. The completed cn^3 sums are infinite and absolutely convergent under $\mathcal{R}$; only the compensation subsets are finite. The principal Mellin exponent is

$$C_b(s) = s + l_x/2 - 1 + h/6 = s - 2/3 - b/6. \quad (8)$$

Here the raw kernel's z coefficient is $h - e$, and the exact tuple adds $e/6$ at the residue. The combined coefficient is $h/6$; it is not charged a second $e/6$.

Proposition 4.1. *For the parameters (3), under the generic low inputs of $\mathcal{R}$,*

$$|I_{\eta, \text{mod}}(Z)| \ll_{\eta, \text{fixed}, \varepsilon} Z^{(1-e)/4 - b/6 + \varepsilon} = Z^{C_b(\sigma_*) + \varepsilon}.$$

Proof. We replay the parameter-independent reflected-energy and Gram calculation of [3]. For a compensation subset write $d_J = \sum_{i \in J} \ell_i$, $e' = e - d_J$, $M' = M - 2d_J$. The exact tuple ratio lies in a fixed positive compact interval; thus $X' \asymp Z^{l_x - d_J}$, $Y' \asymp Z^{l_y - d_J}$ and the common row ball is $Q \asymp Z^{M'}$. Surviving marked slots are summed before the row norm; one common Gaussian profile and Fourier density is retained.

For the actual squarefree dyad H and frozen powerful length O , the sector conditions are $\Delta_H = M' - O - H \geq -O(1/\log Z)$, $2A_0 \leq O + O(1/\log Z)$, $N_0 \leq A_0$ and $z_a \leq e'$. The dual length is $T_d = 2H + 2A_0 + 2z_a - 1 - e' - \theta_N - N_0 - 3B_0$. In particular $T_d - (H - 3d_J) \leq O(1/\log Z) + |\theta_N|$. The generic reflection bound uses $s_{\text{hyb}} = \min\{v, z_a, (v + z_a)/3\}$. On its $s_{\text{hyb}} = z_a$ branch the exponent is at most $M' - \Delta_H$; on the other branch $s_{\text{hyb}} \geq v/2$. The frozen-part saving is the exact identity

$$O/2 + \Delta_H + S_0 + B_0 + T_d/4 = \frac{2M' + 2z_a - 1 - e' + 2\Delta_H - \theta_N}{4} + \frac{2A_0 - N_0 + B_0 + 4S_0}{4}.$$

The second line is nonnegative. Keeping the kernel saving before squaring and counting frozen parts once gives the complete row estimate

$$\sum_{q_m \ll Q} |B_m^J|^2 \ll Z^{M' + (5e - 1 + d_J)_+ / 4 + \varepsilon}.$$

All far Gaussian annuli are first paid by the full absolute input. No step depends on κ or discards a shared prime or empty slot.

The new parameters satisfy $1/6 < e < 1/5$, $b > 0$, $M' \geq 1 - 3e > 0$ and $l_x - d_J, l_y - d_J > 0$. Consequently $Q, Y' \geq 1$ eventually in fixed polynomial ranges. The source parameter $P_a = Y'^2/Q \asymp Z^b$ is eventually at least 1. Use the complete Gram bound

$$\sum_{q_m \ll Q} |A_m(Y')|^2 \ll (Q/Y')(1 + P_a^{1/6} + P_a^2/Y')Z^\varepsilon.$$

Cauchy with the original separation factor $Q^{-1/2}$ gives a fixed tuple exponent

$$(X')^{1/2} Z^{\frac{1}{2} \max\{0, b/6, 2b-l_y+d_J\} + (5e-1+d_J)_+ / 8 + \varepsilon}.$$

The tuple count, $q_{p_J}^{-3/2}$ and shortened $X'^{1/2}$ together contribute $-d_J$. Relative to $l_x/2 + b/12$, the additional exponent is

$$F_{\text{low}}(d) = -d + \frac{1}{2}(11b/6 - l_y + d)_+ + \frac{1}{8}(5e - 1 + d)_+.$$

Its every slope is at most $-3/8$, so the maximum is at 0. The rational interval checks below give the stronger all-subset gap $l_y - e - 11b/6 > 2/25$. Therefore the complete exponent is $l_x/2 + b/12 = (1 - e)/4 - b/6$. All small reflection, separation, height and logarithmic losses are absorbed in the prescribed ε in the original order. Finally (3) and (8) give the equality. $\square$

5 A continuous certificate at the reference parameter

Only in this section set $\kappa = \kappa_*$ as an algebraic reference. It is not an analytic moment assumption when $\beta_* > \sigma_*$. Writing $y = 1/2 - x$ transforms (5) into

$$D = (5/2 - c) + (1 + 2c)y, \quad P = (1 - c/2) + 2y + 2cy^2.$$

Put

$$K_0 = -1/4 + 5e/4 + b/6, \quad W = 1/2 + 3e/2 - ey,$$

and define the reference endpoint on the actual rectangle by

$$E_* = K_0 + (1/2 + e)\delta + e\delta x - h(1 - R_{*,\kappa_*}) = K_0 + W\delta - h(1 - R_{*,\kappa_*}).$$

Direct expansion, retaining the full whole-slot mean gain, gives

$$-2JE_* = F = A(y)\delta^2 - B(y)\delta + C(y), \tag{9}$$

where

$$\begin{aligned} A &= -2(P - D)(W - h) + hP, \\ B &= 2K_0(P - D) + \frac{5}{3}D(W - h) + \frac{5}{6}hP, \\ C &= -\frac{5}{3}DK_0. \end{aligned} \tag{10}$$

The cleared polynomial F is defined for arbitrary real δ ; the identity with E_* is used only where $J \neq 0$.

Let $Q = 4AC - B^2 = \sum_{j=0}^4 Q_j y^j$. For general e, b, κ , Q_0 is a strictly concave quadratic in b , its b^2 coefficient being $-(3312\kappa^2 - 936\kappa + 67)/(1728\kappa^2)$. The numerator has negative discriminant and positive leading coefficient. Its maximizer is

$$b_{\text{opt}} = \frac{5040e\kappa^2 - 900e\kappa + 25e - 432\kappa^2 + 36\kappa + 3}{3312\kappa^2 - 936\kappa + 67}. \tag{11}$$

The maximum is

$$-\frac{25(6\kappa - 1)^2(15\kappa - 2)\{(1314\kappa - 159)e^2 - (36\kappa + 6)e - 30\kappa + 5\}}{162\kappa^2(3312\kappa^2 - 936\kappa + 67)}.$$

At $\kappa = 5/6 - e/2$, the braces are $-p(e)$. Reducing (11) modulo (1) gives exactly the b in (3). Hence $Q_0 = 0$.

All remaining signs have rational certificates. Write $A = \sum_0^3 A_j y^j$ and $C = C_0 + C_1 y$. Table 2 gives strict rational enclosures, obtained by reducing the coefficients in $\mathbb{Q}[e]/(p)$ and applying interval Horner to (2). The endpoints are rational numbers, not decimal evidence.

Coefficient	Strict enclosure	Coefficient	Strict enclosure
A_0	$(47/100, 48/100)$	C_0	$(7/100, 8/100)$
A_1	$(121/100, 122/100)$	C_1	$(6/100, 7/100)$
A_2	$(86/100, 87/100)$	Q_1	$(4/100, 5/100)$
A_3	$(29/100, 30/100)$	Q_2	$(19/100, 20/100)$
		Q_3	$(26/100, 27/100)$
		Q_4	$(7/100, 8/100)$

Table 2: Positive coefficient certificate; $Q_0 = 0$ exactly.

Proposition 5.1. *On $1/50 \leq \delta \leq 3/4$, $0 \leq x \leq 1/2$, the reference endpoint satisfies $E_* \leq 0$. Its unique equality is*

$$y = 0, \quad \delta_* = \frac{5 - 9e}{6 + 18e}, \quad R_{*, \kappa_*} = 2/3.$$

Proof. For every $y \geq 0$, Table 2 implies $A > 0$ and $Q \geq 0$, with $Q = 0$ only at $y = 0$. Thus for arbitrary real δ the cleared polynomial obeys

$$F = A \left(\delta - \frac{B}{2A} \right)^2 + \frac{Q}{4A} \geq 0.$$

On the actual rectangle $J > 0$, so (9) gives $E_* \leq 0$. At $y = 0$, reduction modulo p gives $B(0)/(2A(0)) = (5 - 9e)/(6 + 18e)$ and $R = 2/3$. The same rational bracket gives $1/50 < \delta_* < 3/4$. This proves the continuous assertion without inferring it from finite samples. $\square$

6 The actual parameter and its feedback cost

Assume for contradiction $\beta_* > \sigma_*$ and put $\Delta = \beta_* - \sigma_* > 0$. The bootstrap gives $\Delta \leq (e - 1/6)/4$. The analytic calls to Proposition 2.1 must use

$$\kappa_{\text{act}} = 2\beta_* - 1 \in (\kappa_*, 3/4] \subset [37/50, 1], \quad \kappa_{\text{act}} - \kappa_* = 2\Delta. \quad (12)$$

Then $\beta_* = (1 + \kappa_{\text{act}})/2$ is the true prime-bound prerequisite. Calling the lemma at κ_* instead would assume the conclusion to be proved.

Lemma 6.1. *Uniformly on the count rectangle and $37/50 \leq \kappa \leq 1$,*

$$0 \leq \partial_{\kappa} R_{*, \kappa} = \frac{(\alpha - \delta)^2 \delta x (1 - x)^2}{3\kappa^2 J^2} \leq \frac{625}{36963} < \frac{1}{50}.$$

Proof. Differentiate P/D first: $\partial_c(P/D) = -2x(1 - x)^2/D^2$. Then differentiate (6) and use $\partial_{\kappa} c = -1/(3\kappa^2)$. For the bound use $J \geq (\alpha - \delta)D$, $D \geq 2$, $\delta \leq 3/4$ and $x(1 - x)^2 \leq 4/27$. The resulting bound is $(3/4)(4/27)/(12(37/50)^2) = 625/36963$. $\square$

For the same endpoint geometry, comparison to the actual $C_b(\beta_*)$ gives exactly

$$E_{\beta_*}(h) = E_*(h) - \Delta + h\{R_{*, \kappa_{\text{act}}} - R_{*, \kappa_*}\} \leq -24\Delta/25.$$

Here $h < 1$. The same-source full selected-row expression is

$$\begin{aligned} E_{\beta_*}(d) &= K_{\beta_*} + (1/2 + e)\delta + eq - h(1 - R) \\ &\quad + (d - h)(R + \delta/2 - 17/50), \quad K_{\beta_*} = 2/3 + e + b/6 - \beta_*, \end{aligned} \quad (13)$$

with $R = R_{*, \kappa_{\text{act}}}$. It is the expansion of $a - \beta_* + h(17/50 - 1/6) - al_y - e/2 + eq + d(R + \delta/2 - 17/50)$. Strict local labels and numerator estimates consume the same conductor deficit once; the gain is

still eq . On $1/2 \leq d \leq h$ its slope is at least $33/50 - \delta/2 \geq 57/200$. Hence the endpoint controls all these physical lengths. Choose $\zeta = \Delta/32$; the slope is less than 2, so $h < d \leq h + \zeta$ costs at most $\Delta/16$. The central saving before other real losses is therefore

$$(24/25 - 1/16)\Delta = 359\Delta/400. \quad (14)$$

The gap Δ and actual parameter are fixed before the mesh, slot count or target character.

7 All physical ranges and strict arithmetic admission

The exact rational bracket and parameter definitions give

$$\begin{aligned} l_y - e - 11b/6 &> 2/25, \quad 5e - h - \zeta > 1/50, \\ h + \zeta &< 1, \\ E_{\text{floor}} &= -6/25 + (32/25)e + b/6 < -1/200, \\ E_{\text{middle}} &\leq -71/600 + (329/400)e - (421/1200)b < -1/50, \\ E_{\text{small}} &= (13/75)h - l_y/2 + 63/5000 < -2/25. \end{aligned} \quad (15)$$

These inequalities are recalculated at the cubic parameters, not inherited numerically from [3]. The bound $\zeta \leq (e - 1/6)/128 < 1/384000$ also follows exactly.

The floor $a = 51/100$ uses $\delta = 1/50$, $R = 1$, $q \leq 1/100$ without manufacturing a witness. On $1/100 \leq d \leq 1/2$ the original no-slot count is $R = 76/75 - 2\delta/3$. Its endpoint value is $-73/300 + (13/50)e - (49/300)b + \delta(1/6 + 3e/4 - b/4)$. The coefficient of δ is positive, so $\delta = 3/4$ gives the middle bound in (16). The small bound is already relative to $C_b(\beta_*)$; no second Δ is deducted from it.

Equation (15) gives $e/(h + \zeta) > 1/5$. It exceeds both actual requests in Section 3 by fixed margins. Only $d \geq 1/2$ uses selected primes; each physical length $w_i = \ell_i/d \leq 2\ell_i$. Choose count and moment losses, fixed capacity decrement, uniform mesh and rounding first; then take even K sufficiently large that all w_i fit that mesh and the whole-slot rounding loss is small. Amplitude and bin widths follow. Retain the original finite Θ coefficient combinations, natural row zeros and disjoint physical and internal prime pools. All factors and witnesses are conjugated together on the plain side; separated heights retain their common Sobolev costs.

Outside- S ramification gives the original inducing exclusions. Remaining S -supported sixth-power-free rows are finite and are covered by the bounded-row argument, not by a false generic moment call. The cumulative $T_1/2$ and all original profile and height fees are retained. Thus (6) is an actual count with strict widths, not merely a nominal interpolation.

8 Euler domains, the principal term and outer rows

The parameter change does not alter a local coefficient, zero extension or compensation operation. Retain the raw weight

$$X^{1/2-z} Z^{s+z-1} Y^{w-1} e^{(s+z-1)^2} M(z) \widehat{W}_1(w)$$

and the complete selected-prime sum with its coefficientwise G_p factors and normally convergent unselected correction. There is no division by a general local correction H_p . The quotient is used only in the principal and dynamic domains where its near-one condition has been proved.

On the principal box $\text{Re } s \geq \sigma_*$, $\text{Re } w \geq 19/20$, $\text{Re } z \geq 33/200$, the original coefficient tables give

$$\begin{aligned} c_{\text{good}} &= \min\{6\sigma_* - 401/100, \sigma_* - 3/50, 47/50\} = \sigma_* - 3/50 > 81/100, \\ c_{\text{ram}} &= \min\{\sigma_* - 1/20, 3\sigma_* - 3/2\} = \sigma_* - 1/20 > 82/100. \end{aligned}$$

All denominators retain their separation from zero. The general local term $4 - 6s - 6z$ includes $\theta = (-\operatorname{Re} w)_+$; in $D_1(\varepsilon_0)$ use $\varepsilon_H = \min(\varepsilon_0, 1/50)$. These are full normal-convergence and height domains. Whole-bin contours are moved globally before the original buffered amplitude and error partitions.

For the principal row retain exactly

$$H_\eta(s) = \prod_{p \notin S} H_p(s, 1, 1/6).$$

A target-independent P_0 from the positive majorant gives $|H_\eta - 1| \leq 1/2$ for $\operatorname{Re} s \geq \sigma_*$. The final target-dependent enlargement of S only decreases that majorant; it is the same S in all physical and principal expressions. For the original fixed-ray windows,

$$S_i(Z) = \sum_{p \in \mathcal{P}_i(Z)} W_i(q_p/P_i) q_p^{-5/6} \sim c_i P_i^{1/6} / \log P_i, \quad c_i > 0,$$

$$A_T = (-1)^K Z^{-e/6} \prod_i S_i(Z), \quad c_S = \widehat{W}_1(1) M(1/6) (\operatorname{Res}_{v=1} \zeta_F^S(v))^2 / 6 > 0.$$

Thus A_T is eventually nonzero and $|A_T| \asymp (\log Z)^{-K}$, with subpower inverse. No new normalizer is introduced.

The global principal move, residues at $w = 1, z = 1/6$ and remaining lines $19/20, 33/200$ give

$$m_w = l_y/20 > 0, \quad m_z = h/600 > 0, \quad 0 < \mu < \sigma_* \min_i \ell_i.$$

The double-residue s line moves only to the right, to 2. The four local principal error exponents remain $-\operatorname{Re} s, -6\operatorname{Re} z, 4 - 5\operatorname{Re} s - 6\operatorname{Re} z$ and $1 - \operatorname{Re} w - 6\operatorname{Re} z$; their weakest decay at the residue is σ_* .

For small rows $d < 1/100$ and bounded nontrivial units, use the local proof on $D_1(1/3)$ since $\sigma_* + 1/2 > 4/3$; do not invoke a numerically stronger old $D_1(3/8)$ condition. The full numerator and absolute tuple give exactly the small bound (16). For $U > Z^{h+\zeta}$ use the full completed absolute sum on $(2, 2, z_\infty)$, whose exponent is

$$B_0 + (h + \zeta)(1 + \varepsilon) - \zeta z_\infty + \varepsilon, \quad B_0 = l_x/2 + 1 + l_y = 7/4 - 3e/4 + b/4.$$

After ζ is fixed, select sufficiently large fixed z_∞ before the target, then the small losses and internal height orders. This pays any required outer saving. An order is not changed with a moving row or with Z .

9 Uniform choices and the family contradiction

Geometry, the bootstrap, Δ and κ_{act} precede all target choices. Allocate small fixed fractions of Δ to count and moment losses, capacity decrement, mesh and rounding. Then fix even K and all windows, followed by $\mu = \sigma_* e / (2K) > 0$, amplitude widths, prime losses and buffers. Keep total other central costs below $\Delta/4$ in (14). Floor, middle and small losses retain at least half their fixed margins; principal losses are separately smaller than $m_w/4, m_z/4, \mu/4$. Fix ζ, z_∞ in this same target-independent phase. This order avoids choosing the mesh or K backwards from a K -dependent μ .

Set

$$m_0 = \min\{\Delta, m_w, m_z, \mu, 1/200, 1/50, 2/25\}, \quad m = m_0/4.$$

For a target η , fix its arithmetic datum, the final S and then the finite internal Sobolev, seminorm and height orders. For all sufficiently large Z define the physical function and for all positive Z define the principal function by

$$J_\eta(Z) = I_{\eta, \text{mod}}(Z) / (c_S A_T(Z)), \quad f_\eta(Z) = \frac{1}{2\pi i} \int_{(2)} Z^{C_b(s)} e^{(s-5/6)^2} \frac{H_\eta(s)}{L_F^S(s, \eta)} ds.$$

Both are independent of the auxiliary cutoff T_1 . Take the total low and inverse-normalizer loss $\omega = \Delta/2 < \Delta$. Then

$$|J_\eta(Z)| \ll_\eta Z^{C_b(\sigma_*)+\omega}.$$

The common principal and all high estimates give finite A_η, B_η such that, for an external order N ,

$$|J_\eta - f_\eta| \ll_{\eta, N} Z^{C_b(\beta_*)-m}(1+T_1)^{A_\eta} + Z^{B_\eta} T_1^{-N}. \quad (17)$$

The internal orders giving A_η, B_η are fixed before N . For the actual detector ceiling $\tau_{0,\eta} > 0$, choose

$$0 < \tau_\eta \leq \min\{1/10000, \tau_{0,\eta}, m/[4(A_\eta + 1)]\}, \quad T_1 = Z^{\tau_\eta}.$$

Choose N_η afterwards so that $B_\eta - N_\eta \tau_\eta < C_b(\beta_*) - m/2$, and then the eventual threshold in Z . The common high saving is $\sigma_{\text{hi}} = m_0/8 > 0$. Target-dependent constants, orders and thresholds do not change it.

Proof of Theorem 1.2. The low and high bounds imply local uniform convergence of

$$\mathcal{F}_\eta(s) = \int_0^\infty f_\eta(Z) Z^{-C_b(s)} \frac{dZ}{Z} \quad \text{on } \operatorname{Re} s > \beta_* - \varepsilon_*, \quad \varepsilon_* = \min\{\Delta/2, m_0/8\} > 0.$$

At small Z use only f_η , whose Gaussian integral has rapid decay by rightward shifts. There is no need to define J_η or divide by A_T there. At large Z use (17) and the strict low gap. Since C_b has slope 1, the same Mellin/Fourier identification as in $\mathcal{R}$, initially on line 2, gives

$$\mathcal{F}_\eta(s) = e^{(s-5/6)^2} H_\eta(s) / L_F^S(s, \eta).$$

Analytic continuation makes this reciprocal expression holomorphic on the larger half-plane, without moving a contour across a target zero. Since $\beta_* - \varepsilon_* > \sigma_*$ and $|H_\eta - 1| \leq 1/2$ there, a zero cannot be canceled by the numerator.

The supremum need not be attained. Its definition supplies a target zero with real part $> \beta_* - \varepsilon_*$. All common choices preceded that target, and the holomorphic reciprocal contradicts its existence. Thus $\beta_* \leq \sigma_*$. Finite deleted Euler factors have no zero in $\operatorname{Re} s > 0$, so imprimitive functions have the same strict conclusion. The quadratic factorization in $\mathcal{R}$ transfers it to the Dirichlet family. At 1 the principal pole is allowed; the zeta factor uses $L(1, \chi_{-3}) = \pi/(3\sqrt{3}) \neq 0$. $\square$

10 Exact comparison and scope of the advance

The preceding free- b paper gives $\sigma_0 = (1507 - 2\sqrt{921})/1653$. Its associated e_0 is the positive root of $1653e_0^2 - 66e_0 - 35 = 0$, isolated by

$$\frac{16683838746}{10^{11}} < e_0 < \frac{16683838748}{10^{11}}.$$

The lower endpoint for e in (2) exceeds this upper endpoint. Hence $e > e_0$, and exactly $\sigma_* < \sigma_0$. For orientation, $\sigma_0 - \sigma_* = 0.000000050379069793 \dots$. This comparison is not used as a bootstrap.

The advance is the re-established plain contract below $3/4$ and its actual count feedback. The theorem remains relative to $\mathcal{R}$. It does not independently certify the source paper, its analytic inputs or an external proof kernel, and does not establish RH. The separate critical-line proportion route still needs a complete signed response and finite-constant fourth-moment budget; none is supplied by this strip gain. The equality of the reference envelope is a statement about a count certificate, not a constructed bad row or zero.

A Reproducible exact certificate

The file `scripts/hybrid_kappa_feedback_exact_audit.py` uses only rational arithmetic. It implements $\mathbb{Q}[e]/(p)$ in the basis $(1, e, e^2)$, with $e^3 = (-20 - 21e + 954e^2)/657$, and solves the exact three-dimensional multiplication system for division. Modulo 7, $p(j)$ for $j = 0, \dots, 6$ has residues 6, 3, 4, 3, 1, 6, 5; the cubic is irreducible. Endpoint signs and $p' < 0$ on $(1/6, 167/1000)$ certify the selected real embedding.

For a direct way to reproduce Table 2, put $D_0 = 5/2 - c$, $D_1 = 1 + 2c$, $P_0 = 1 - c/2$ and $T_0 = P_0 - D_0$, $T_1 = 2 - D_1$. Since $W - h = -b/2 - ey$,

$$\begin{aligned} A_0 &= bT_0 + hP_0, & A_1 &= bT_1 + 2eT_0 + 2h, \\ A_2 &= 2bc + 2eT_1 + 2hc, & A_3 &= 4ec, \\ B_0 &= 2K_0T_0 - 5bD_0/6 + 5hP_0/6, \\ B_1 &= 2K_0T_1 - 5bD_1/6 - 5eD_0/3 + 5h/3, \\ B_2 &= 4K_0c - 5eD_1/3 + 5hc/3, \\ C_0 &= -5K_0D_0/3, & C_1 &= -5K_0D_1/3. \end{aligned}$$

Then $Q_j = 4 \sum_{r+s=j} A_r C_s - \sum_{r+s=j} B_r B_s$, with absent coefficients zero. Reduce these expressions modulo p and use interval Horner on (2). The strict rational enclosures in the table follow; they were also independently verified with a rational Bernstein enclosure. The complete rational coefficient triples are saved in `output/hybrid-kappa-feedback-exact-audit.json`.

Run from the repository root:

```
python scripts/hybrid_kappa_feedback_exact_audit.py
```

The audit has 149 counted checks, including 98 identity checks at 49 finite endpoint and square-completion models. It also contains uncounted internal arithmetic assertions. The continuous proof is coefficient positivity and square completion, not those 49 samples. Its scope is displayed exact algebra, root isolation, strict parameter inequalities and finite identity models. It does not verify the plain induction, imported analytic inputs, actual prime sums, infinite contours, the zero-free conclusion, RH or an external kernel. Its source-hash field is fixed metadata: the script does not read or authenticate the external source or Note 451. Actual version binding is performed separately by the independent reviews and build manifest.

The lower- κ source replay and full research derivation are preserved in [4, 5, 6]. Two full independent reviews of this paper, its final source and PDF hashes, compilation and all-page visual verification are recorded in `reviews/2026-10-07/kappa-feedback-paper-build-manifest.json`. The reviewed frozen research draft and the independently reviewed paper are separate evidence objects.

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